"INNOVATION IN ENVIRONMENTAL TESTING SOLUTIONS"

Vibration Combined

Climatic Test Chamber

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Vibration Combined Climatic Test Chamber

SESS Vibration Test Chambers are designed to provide reliable and precise solutions for vibration and environmental testing. Our V-ETC-Series and V-TC-Series chambers combine temperature, humidity, and vibration conditions in a single system, enabling comprehensive product validation under real-world stress environments. With rapid temperature change rates and robust engineering, these chambers ensure accurate simulation of extreme conditions to test product performance, safety, and durability.

MECF + TUV LAB, PUNE

MODEL :

Vibro-ETC-2250 / −70°C to +180°C / RH / W / C5 / H5 / PLC

INTERNAL DIMENSIONS :

1500 mm (W) × 1500 mm (D) × 1000 mm (H)

RAMP RATE : 5 °C/MIN

MAEON LAB - PUNE

MODEL :

Vibro-ETC-2250 / −70°C to +180°C / RH / W / C10 / H10 / PLC

INTERNAL DIMENSIONS :

1500 mm (W) × 1500 mm (D) × 1000 mm (H)

RAMP RATE : 10 °C/MIN

AANCHAL LAB - BANGALORE

MODEL :

Vibro-ETC-1700 / −70°C to +180°C / RH / W / C10 / H10 / PLC

INTERNAL DIMENSIONS :

1200 mm (W) × 1200 mm (D) × 1200 mm (H)

RAMP RATE : 10 °C/MIN

PERFORMANCE :

The chamber is designed to perform both steady-state tests (maintaining a constant temperature/humidity over long durations) and cyclic tests (varying temperature and humidity in programmable ramp/soak profiles). This ensures compliance with multiple international testing standards such as IEC, ISO, and MIL-STD for environmental stress testing of components, materials, and assemblies.

CHAMBER TYPE

VIBRO-ETC : VIBRATION COMBINED CLIMATIC TEST CHAMBER :

This chamber is specially designed to integrate environmental stress testing with vibration testing. It allows simultaneous application of temperature, humidity, and vibration stresses, enabling accurate simulation of real-world operating environments for aerospace, defence, automotive, and electronic products.

WORK SPACE

4500 LITERS – 2000 MM (W) × 2000 MM (D) × 1125 MM (H)

- Provides a large usable volume for mounting components and test specimens.
- The chamber is dimensioned to accommodate fixtures that integrate with electrodynamic vibration shaker systems.
- Internal usable space is optimized to ensure uniform airflow and thermal distribution.

TEMPERATURE RANGE

-70 °C TO +180 °C ±1 °C (RTD PT100 CLASS A SENSOR), TEMPERATURE GRADIENT ≤ 2 °C

- Achieves wide temperature extremes for high/low thermal stress.
- The Class A PT100 RTD sensor ensures high accuracy (±0.15 °C at 25 °C).
- Meets stringent testing requirements for thermal cycling, thermal shock, and accelerated aging.

TEMPERATURE RANGE

10% to 98% RH, Humidity fluctuation ±1% to 3% (within +10 °C to +85 °C range, electronic sensor Rotronic make)

- Provides climatic humidity testing across a wide range.
- Rotronic sensor is an internationally recognized Swiss make, offering high accuracy and long-term stability.
- Enables damp heat testing, accelerated corrosion testing, and electronics moisture reliability tests.

REFRIGERATION UNIT – CONDENSER TYPE WATER-COOLED CONDENSER WITH SUITABLE CHILLER

- SESS will supply a dedicated chiller unit, connected with CPVC pipeline (10 m length).
- Water-cooled systems offer higher efficiency and stable cooling compared to air-cooled types.

RATE OF CHANGE OF TEMPERATURE :

Cooling Ramp: 5 °C/min (empty chamber) as per IEC 60068-3-5, averaged between +180 °C to -70 °C with sensor at blower discharge.

H5 Heating Ramp: 5 °C/min (empty chamber) as per IEC 60068-3-5, averaged between -70 °C to +180 °C.

These ramp rates ensure fast thermal cycling capability, critical for stress screening testing. Compliance with IEC ensures internationally accepted test reproducibility.

4.5.2 Temperature rate of change

To determine temperature rate of change:

- adjust chamber to lowest specified temperature and allow to stabilize;
- adjust chamber to highest specified temperature, monitoring the time between the 10 % and the 90 % points of the temperature range;
- allow chamber to stabilize at the highest specified temperature;
- adjust chamber to lowest specified temperature, monitoring the time between the 90 % and 10 % points of temperature range.

This will determine the temperature rates of change for heating and cooling in K/min.

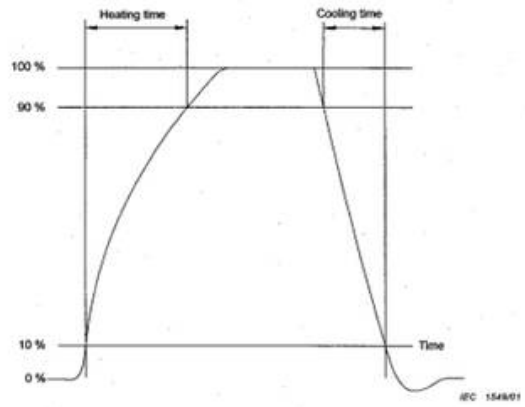


Figure 5 – Temperature rate of change for heating and cooling of a test chamber

CHAMBER CONTROL TYPE

SIEMENS S7-1200 PLC WITH 7" WEINTEK HMI :

- Industrial-grade Siemens PLC ensures high reliability and fast control response.
- Weintek 7" touchscreen HMI provides user-friendly graphical interface for programming and monitoring.

FEATURES INCLUDE:

- Profile programming (multi-segment ramp/soak)
- Real-time trend display
- Alarm logs and diagnostics
- Remote monitoring and data logging (Ethernet/Modbus).

VIBRATION TABLE CONNECTING THERMAL BAG

- Vertical and horizontal axis thermal bag provide flexible connectivity between the climatic chamber and vibration system.
- Designed with screw rod and motorized up/down movement, ensuring quick, leak-free sealing during chamber-shaker integration.
- Maintains thermal insulation while allowing vibration transfer.

CHAMBER VIBRATION TABLE CONNECTIVITY FIXTURE

- Chamber bottom lid cut-out: 1500 mm × 1500 mm – compatible with horizontal vibration shaker table.
- Ensures precise alignment and rigid sealing, allowing test specimen to experience combined environmental & vibrational stress simultaneously.

CHAMBER VIBRATION TABLE DUMMY DOOR

- One dummy lid provided to cover the shaker opening when vibration testing is not in use.
- Maintains chamber integrity and allows full standalone climatic testing without vibration system.

DOOR :

- Full front opening door, double-walled insulated construction.
- Inspection glass window (400 × 400 mm) with heated anti-condensation design for clear viewing during test.
- Provides easy specimen loading/unloading with secure insulation.

GASKET :

- Double gasket sealing arrangement – one on chamber body, one on door.
- Ensures leak-proof closure, minimizing heat/cool loss and maintaining uniform temperature/humidity inside the chamber.

HINGES & LATCHES :

- Heavy-duty stainless-steel hinges for long life and durability.
- Toggle-type locking latches provide tight closure and even pressure distribution across the gasket, ensuring maximum sealing efficiency.

REFRIGERATION SYSTEM :

- Type: Semi-hermetic compressor, two-stage cascade water-cooled system.
- The two-stage cascade design enables achieving very low temperatures (−70 °C) with high stability.
- Semi-hermetic compressor ensures durability, ease of service, and long operational life.
- Water-cooled condenser system increases energy efficiency and ensures reliable performance even in high-ambient conditions.
- Designed as per international safety and performance standards for environmental test chambers

CHAMBER CONSTRUCTION

The chamber is designed with a double-walled, insulated structure to ensure superior thermal efficiency and durability.

Inner Chamber: Fabricated from TIG-welded stainless steel (SS 304) sheets of 1.5 mm / 1.2 mm thickness, thoroughly tested for leakage using both water and air pressure methods.

Outer Chamber: Constructed from CRCA or GI sheets (as per customer's tender RFQ), with 1.5 mm thickness, ensuring structural strength. The outer body is precision-manufactured using CNC technology for smooth, seamless joints without protrusions.

SURFACE FINISH :

- Door: Epoxy-based powder coating in RAL 9001 Cream White.
- Main Body: Powder coated in RAL 7024 Graphite Grey.

DOOR DESIGN :

- Configuration: Fully frontal opening, double-walled insulated door with a vertical trolley mounting arrangement.
- Sealing: High-quality silicone rubber gasket (exclusively sourced) ensures optimal air-tight sealing.
- Hardware: Equipped with robust hinges and a toggle-type locking mechanism for secure closure and reliable insulation.

This meticulous construction provides strength, precision, and long service life, while also ensuring a clean and aesthetic appearance suitable for advanced laboratory and industrial applications.

INTERIOR CONSTRUCTION :

- Material: Stainless Steel SS304, thickness 1.5 mm.
- Fabricated using TIG welding for leak-proof, corrosion-resistant construction.
- Smooth welded joints prevent condensation and ensure long-term chamber durability.
- SS304 offers excellent resistance to corrosion, chemicals, and humidity cycles, making it ideal for long-term environmental testing.

EXTERIOR CONSTRUCTION :

- Material: Cold Rolled Close Annealed (CRCA) steel sheet, thickness 1.5 mm.
- Powder-coated finish provides enhanced protection against rust, scratches, and environmental degradation.
- CRCA construction ensures high mechanical strength and rigid support for chamber operations.

INSULATION :

- Thickness: 150 mm high-density glass wool insulation.
- Provides superior thermal resistance, minimizing heat loss during low-temperature operation and reducing power consumption.
- Ensures uniform internal temperature with minimal fluctuations.

PORT HOLE :

- Size/Quantity: 50 & 100 mm diameter – 1 each
- Used for introducing electrical cables, sensor probes, or additional test fixtures into the chamber without disturbing thermal sealing.
- Fitted with removable silicone plugs to maintain airtight integrity when not in use.

INNER LAMP :

- Illumination: 50 W – 1 No.
- Interior test space fitted with moisture-protected lamp for specimen visibility.
- Provides clear view during tests, even at low temperatures and high humidity conditions

DOOR CONSTRUCTION :

Quantity: 2 No., full front opening.

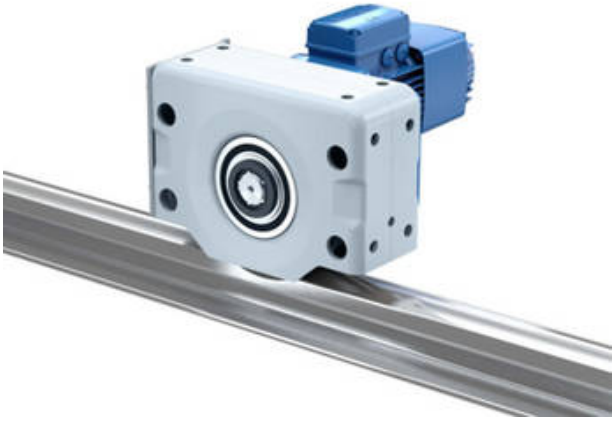
Design: Double-walled insulated with stainless steel SS304 inner lining.

Allows easy access for specimen loading/unloading, while ensuring maximum sealing efficiency.

MOUNTING & MOVEMENT / INTEGRATION WITH VIBRATION SYSTEM

MOTORIZED CHAMBER MOVEMENT :

The chamber/test space unit shall be equipped with motorized movement to enable precise integration with the vibration shaker's armature and slip table. This feature will be supplyas part of the system. Motorized travel ensures effortless positioning, minimizes manual effort, and provides consistent alignment during coupling operations.



TEMPERATURE MEMBRANE CLOTH BARRIER :

A temperature-resistant membrane cloth barrier (or equivalent sealing arrangement) will be provided at the interface points for Armature/Head Expander integration and Slip Table bottom panel integration.

- This barrier prevents thermal leakage between the vibration system and climatic chamber.
- Maintains temperature and humidity uniformity during combined environmental and vibration testing.

TRACK CONSTRUCTION :

The chamber is mounted on heavy-duty manual tracking wheels running on a track fabricated from 5 mm thick square tubular steel, ensuring high structural rigidity and long-term durability.

TRACK LENGTH :

A 7.5-meter track is provided, offering sufficient travel distance between:

- Test position (coupled with the shaker system)
- Standby/service position (for chamber maintenance or non-vibration use).

STOPPER ARRANGEMENT :

Heavy-duty mechanical stoppers are installed at both ends of the track to ensure:

- Safe stopping, Positional accuracy, Stability of the chamber during testing operations.

RELIABILITY :

The robust design guarantees safe, repeatable positioning of the chamber when integrated with electrodynamic shakers for combined vibration–climatic tests,

OPERATIONAL ADVANTAGE :

This combined motorized/manual tracking design ensures smooth, controlled chamber movement, enabling quick transition between operational and service modes without misalignment.

CHAMBER OPERATION MODES :

- Operation Modes: Constant and Program
- Program Mode Settings: Pause/start condition, program name, and list

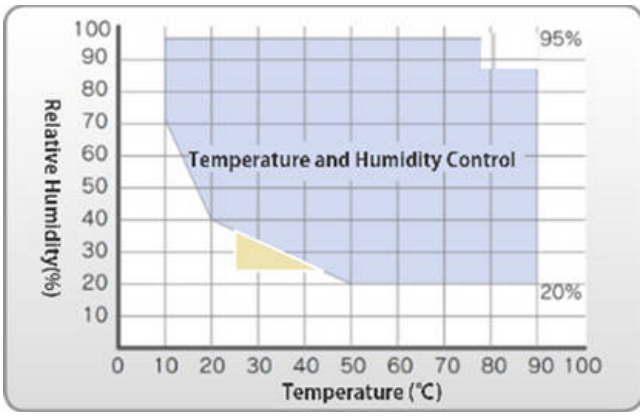
HUMIDITY RANGE :

Humidity Range: 10% to 98% RH (for the range of 10°C to 85°C)

Control Accuracy: ± 2% to 3% RH

This humidity control feature makes it ideal for tests that require precise moisture conditions.

Condenser Type: water-cooled, 2 Stage refrigeration system.



IN OUR CLIMATIC TEST CHAMBER'S AIR CIRCULATION SYSTEM, WE INCORPORATE THE FOLLOWING COMPONENTS AND FEATURES:

Fan/Blower: The chamber is equipped with a fan or blower featuring a continuous-duty rated motor. This motor efficiently circulates air within the chamber.

Conditioning Space: Inside the main chamber, there's a dedicated conditioning area with baffles, housing essential components like heaters and cooling coils. These elements play a crucial role in air conditioning.

Uniform Temperature: Conditioned air is introduced into the main chamber to ensure consistent temperature distribution throughout the workspace, maintaining a stable temperature environment during tests.

Laminar Airflow: The chamber is designed for laminar airflow, achieved using a fan motor and impeller with an appropriate capacity. This setup guarantees even distribution of temperature and humidity within the chamber, adhering to specified limits.

Motor Shaft: TC/ETC - The motor shaft used for chamber air circulation is constructed with SS 304 without any joints. The motor is positioned outside the chamber, while an extended-length SS 304 shaft drives the fans.

ATC/AETC - "The mechanical design of the chamber's air circulation motor is constructed with a sturdy shaft connected by a flexible coupling. The inner shaft is made of SS 304 material and is connected to the blower drum, which features a double vacuum seal mechanism. The motor is positioned outside the chamber, and it drives the blower drum through the inner SS 304 shaft. This design ensures smooth and reliable operation while maintaining the integrity of the chamber's internal environment."

Motor Winding: For high-temperature chambers, we utilize motor windings of H class, whereas for chambers operating below 130°C, F class motor windings are employed.

These features combine to create an efficient and precise air circulation system within the climatic test chamber, facilitating accurate and dependable test conditions for various experiments and applications.

INCONEL-TYPE AIR HEATERS WITH PTFE WIRE AND SILICON END CAP

Heating System: The chamber is equipped with sealed inconel tube heaters to maintain a uniform temperature. These heaters are meticulously designed to ensure even heat distribution throughout the chamber. Heat input is regulated by solid-state relays, controlled through a PID controller programmer.

One notable advantage of inconel tube heaters is their exceptional ability to withstand high temperatures and corrosive environments. Inconel, known for its outstanding durability and heat resistance, is well-suited for prolonged use in challenging conditions.

Compared to stainless steel heaters, inconel heaters exhibit superior resistance to oxidation, corrosion, and high-temperature degradation. They are engineered to deliver reliable and efficient heat transfer without direct radiation onto the test item, ensuring a uniform heating process that doesn't adversely affect the test specimen.

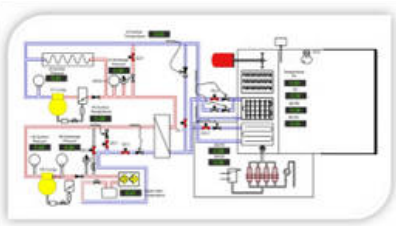
In conclusion, the use of sealed inconel tube heaters is highly recommended for maintaining precise and uniform chamber temperatures while ensuring the long-term performance and durability of the heating system.

REFRIGERATION SYSTEM :

The SESS Vibration Combined Test Chamber is equipped with a robust and reliable cascade refrigeration system, designed for precision performance and long service life.

Compressor:

Semi-hermetic, 2-stage compressors of reputed make (Bitzer / Dorin / Bock), mounted on anti-vibration pads to reduce vibration and noise.



Refrigerants:

CFC-free R404A (high stage) and R23 (low stage), ensuring compliance with environmental standards and reliable ultra-low temperature performance.

Anti-Corrosive Design:

All refrigeration components are manufactured from anti-corrosive materials to provide durability and extended service life.

Oil Return System:

An integrated oil return mechanism prevents oil migration from the compressor. A sight glass is provided for continuous oil level monitoring.

Heat Exchanger Coils:

Made of inner-grooved copper tubes with aluminum fins, engineered for maximum heat transfer efficiency.

Cascade System:

Chamber cooling is achieved through a cascade refrigeration cycle, with R23 operating in the low stage and R404A in the high stage. This ensures the required low-temperature stability and performance.

Component Arrangement:

All refrigeration components are housed in a dedicated compartment, with adequate ventilation to prevent excessive heat build-up.

This refrigeration system design ensures efficient, reliable, and compliant performance of the climatic vibration test chamber.

REFRIGERATION SYSTEM :

BITZER MAKE SHELL & TUBE WATER-COOLED CONDENSER

The SESS Vibration Combined Test Chamber can also be equipped with shell & tube water-cooled condensers, offering a standard and robust solution for all normal and technical cooling applications.

Refrigerant Compatibility:

Suitable for all commonly used refrigerants including HFC, HFO, and HFC/HFO blends. A special version is also available

Construction & Reliability:

Unique tube-brazing process and coated tube sheets guarantee mechanical strength and long service life.

Low-fouling tube design minimizes scaling and contamination, enabling the condenser to deliver consistent and efficient performance throughout its lifetime.

This condenser option provides a reliable, durable, and efficient heat rejection system, making it well-suited for a wide range of industrial and laboratory test chamber applications.

SHELL & TUBE WATER-COOLED CONDENSER SYSTEM :

The SESS Vibration Combined Test Chamber is designed with a shell & tube water-cooled condenser system on the chamber side, where R404A refrigerant rejects heat to chilled water. On the chiller side, a stainless steel brazed plate heat exchanger (PHE) is utilized to ensure high thermal efficiency and reliable long-term performance.

Condenser Construction :

- Shell & Tube Heat Exchanger on the chamber side (R404A to chilled water).
- Stainless Steel Brazed Plate Heat Exchanger (PHE) on the chiller side, designed to withstand specified working pressures on both the waterside and refrigerant side.

Accessories Supplied :

All required accessories—including chiller unit, pipelines, and circulation pump—will be listed and supplied to ensure seamless operation of the condenser and chamber within the specified working range.

Installation Flexibility :

The system is engineered for installation with up to 10 meters separation between the chiller unit and the test chamber, providing layout flexibility and ease of integration at the customer site.

This configuration fully complies with the requirement for a water-cooled condenser system, combining shell & tube and brazed plate heat exchanger technologies, and supplied as a complete accessory package for dependable and efficient performance.

CONTROLLER FEATURES :

PLC-Based Control System: Siemens S7-1200

HMI: 7-inch Weintek Touchscreen Interface

Internal Memory: 4 GB

RAM: 1 GB

Data Logging: Supports up to 90 days of continuous data storage

Connectivity Ports:

- USB: 1 Port, Ethernet: 2 Ports, RS485: 1 Port

Additional functions :

- CSV data files and operation log downloadable via USB or Ethernet
- Multi-user login with role-based access (Admin / Engineer / Operator)

Optional PC Interface :

Remote Monitoring & Control – LabVIEW-based software.

Key functions

- Parameter upload & profile editing
- Live trend display and alarm pop-ups
- Automatic data logging and batch download

The PC interface enables full remote operation and data acquisition, integrating easily with laboratory information systems.

WEINTEK 7" HMI – MODEL CMT2078X :

Display: 7" TFT LCD, resolution 800 × 480, 16.7M colors

Brightness / Contrast: 400 cd/ ㎡, contrast ratio 800:1

Backlight: LED type, lifetime >30,000 hrs

Touch Panel: Capacitive type, wide viewing

Memory: 4 GB Flash, 1 GB RAM

Data Storage: USB host port (supports external USB storage >50 GB for data logging)

Networking: Dual Ethernet ports (10/100/1000 Base-T), optional Easy Access 2.0 for remote monitoring

Operating Range: 0 ~ 50 °C, 10–90% RH (non-condensing)

Certifications: CE marked, UL/cULus listed

Mechanical: Panel mounting, IP65 front protection, dimensions 192 × 138 mm

EFFICIENT REMOTE MONITORING – CMT VIEWER

The Weintek cMT Viewer provides efficient remote monitoring with a very fast refresh rate using Weintek's exclusive application.

The app runs independently on multiple devices, making it ideal for multi-party cooperative monitoring and ensuring seamless access to real-time chamber performance data.

MULTIPLE WAYS FOR REMOTE MONITORING

THE CLASSIC VNC

Use VNC viewer to monitor HMI Screen

Synchronously. The Classic Way.

THE CONVENIENT : WEBVIEW

Go app less and monitor with a browser such as Chrome/ Safari.

Best in terms of cross platform compatibility.

THE EFFICIENT CMT VIEWER

Monitor with very fast refresh rate using Weintek's exclusive app.

App run independently perfect for multi-party cooperative monitoring.

INSTRUMENTATION & CONTROL SYSTEM :

The SESS Vibration Combined Test Chamber is equipped with an advanced Instrumentation & Control system, designed for accuracy, reliability, and user-friendly operation.

Wiring:

All flame-retardant cables are used for both main power and control circuits, complying with international and equivalent Indian standards.

Data Handling:

- USB provision for fast data download.
- HMI with 4 GB internal memory, plus provision for external USB storage (>50 GB) to directly save data logging and trend information.

Data Visualization:

- Real-time trend graphs of temperature and humidity vs. time viewable on HMI.
- Instant profile preview in graphical format for quick verification.

HMI / Panel:

Weintek 7" touchscreen HMI, IP65 protected, with intuitive user interface.

Controller / PLC:

- Siemens S7-1200 PLC for reliable automation and process control.
- High-end multi-loop PID controller (Eurotherm or equivalent) for precise simultaneous control of temperature and humidity.

Programming:

User-friendly interface for test programming and sequence creation, ensuring simple and flexible operation.

Monitoring:

- Digital Output (DO) life monitoring integrated into PLC logic to track run-hours of each component.
- Internal fault alerts displayed on HMI for proactive maintenance.

Fault Diagnostics & Alerts:

- All internal faults displayed on HMI with a HELP menu for guided troubleshooting.
- Fault history with diagnostics log stored for future reference and analysis.

Water System Monitoring:

Low water level indication for the humidity system displayed on HMI, ensuring timely refilling and uninterrupted operation.

Power Resumption Modes:

In case of power failure, the system supports:

- Conditional restart (resume based on last temperature/time conditions).
- Continuous restart option for uninterrupted testing.

Chamber–Shaker Interface:

Dedicated interface circuits and interlocks ensure safe and synchronized combined operation of the climatic chamber and vibration shaker system.

HUMIDITY SYSTEM :

Water Requirement: 2 - 4 liters per hour (LPH)

Water Inlet Connection: 12 mm press-fit

Drain Line: 12 mm press-fit connection

Boiler Tank: Immersion heater-based, with an automatic water filling system and electrical water level sensor

This system provides continuous humidity control for prolonged tests.

The boiler system is constructed with a SS 304-grade tank, integrated with a high-quality immersion heater. The heater meets internal safety standards with a maximum leakage current of 30 mA. The boiler also includes an automatic water filling system, using a GEV (Germany make) solenoid valve, along with a water level electronic sensor for reliable and safe operation.

DEHUMIDIFICATION SYSTEM :

Dehumidification: Controlled via PLC logic, utilizing a solenoid valve and cooling coil to reduce chamber humidity. The chamber automatically adjusts humidity levels using a sophisticated dehumidification system integrated with the refrigeration system.

The dehumidification system is controlled via PLC logic. It operates through a de-humidity solenoid valve, integrated with the refrigeration system. The cooling coil contains a refrigeration cold tube that absorbs moisture from the chamber's air. The system monitors the cold tube temperature through the PLC. When dehumidification is required, the cold tube captures the moisture, freezes it into water particles, and drains the condensed water, effectively reducing the chamber's internal humidity.

Condensation Protection and Advanced Dehumidification System

Our test chamber is designed to provide reliable condensation protection and precise humidity control, ensuring optimal testing conditions for sensitive specimens. Key features include:

Condensation Protection with Separate Dehumidifying Coil:

The test chamber is equipped with a separate dehumidifying coil that effectively removes moisture from the air, minimizing the risk of condensation on the test specimen.

This dedicated coil helps maintain the integrity of the testing environment, ensuring accurate and reliable test results without unwanted moisture affecting the specimen.

ADVANCED DEHUMIDIFICATION AND HUMIDITY CONTROL SYSTEM :

RTD Sensor for Tube Skin Temperature Control:

- An RTD sensor is placed on the dehumidifying refrigeration copper tube coil to monitor and maintain the tube's skin temperature.
- The sensor value can be easily adjusted through the HMI interface, allowing users to set and control the desired temperature and dehumidification levels with precision.
- This ensures consistent regulation of humidity levels within the chamber by controlling the refrigerant flow via the solenoid valve.

Solenoid Valve Regulation:

- A solenoid valve controls the refrigerant flow, freezing moisture from the air within the chamber. This regulation maintains the required dehumidification level according to the set values.
- Moisture is efficiently absorbed from the air and drained out of the chamber, ensuring a dry and stable environment for precise testing.
- This system enhances the reliability and accuracy of tests, particularly for sensitive applications where condensation and humidity fluctuations could affect results.

Integrated in HMI:

A dedicated data logging page is provided within the HMI screen, enabling easy monitoring and configuration.

Channels & Flexibility:

- Total 9 universal input channels available.
- Each channel can be configured through HMI software as either:
- RTD (Pt100) for temperature measurement
- Humidity sensor input
- Analog input (4–20 mA / 0–10 VDC) for external transducers.

Sensor Configuration Supplied:

- 8 Nos. RTD Pt100 sensors (temperature measurement range: –100 °C to +300 °C).
- 1 No. RH sensor (operating range: 10% to 100% RH, temperature compatibility: +10 °C to +85 °C).

Logging & Data Handling:

- Continuous multi-channel data logging with time-stamping.
- Test reports can be downloaded directly via USB from the HMI.
- Remote monitoring option available via Ethernet/Remote Viewer for real-time access and supervision.

User Advantage:

- Provides simultaneous multi-point monitoring of chamber environment and test specimen.
- Flexible sensor input allows integration with different measurement requirements.
- Easy retrieval of logged data in digital format for analysis, audit, and certification compliance.

Specimen Protection and Over-Temperature Protection :

- Over-temperature protection according to EN 60519-2: 2006, with a separate sensor, thermal safety class 2.
- Independent adjustable temperature limiter (tmin / tmax), with the sensor installed in the test space, and individually adjustable fixed values.
- Software-controlled temperature limiter (tmin / tmax), with individually adjustable fixed values.
- Test chamber protection with a safety temperature limiter to prevent over-temperature conditions within the test chamber.

INTEGRATION OF CLIMATIC CHAMBER WITH ELECTRODYNAMIC SHAKER :

We confirm that the climatic chamber will be fully integrated with the electrodynamic shaker system. As part of our scope, we will provide and supply all necessary interfacing elements required for reliable operation and sealing. This includes:

- Temperature membrane cloth barrier (flexible sealing arrangement around the shaker head to isolate test environment).
- Fixtures and adaptors for mounting the test specimens and ensuring vibration + climatic test compatibility.
- Sealing arrangements to prevent leakage of temperature and humidity between the chamber and shaker.
- Cabling provisions with proper routing, shielding, and feed-throughs to withstand vibration and climatic cycles.
- Ducting provisions for air circulation and uniform test condition distribution.

All the above items will be considered within the supply scope to ensure smooth integration, compliance with test standards, and ease of operation.

CONSTRUCTION FEATURES :

Interior :

Material: Mirror-finish stainless steel (SS 304)

Thickness: 1.2 / 1.5 mm

Construction: Skillfully TIG-welded for durability TIG-welded for durability

Exterior:

Material: Cold Rolled Close Annealed (CRCA) sheet or SS 304

Thickness: 1.5 / 1.2 mm

Insulation:

Type: 150 mm glass wool

Purpose: Energy efficiency and maintaining stable internal temperatures

DOOR FEATURES :

Type:

Full front-opening, double-walled insulated door with a stainless steel interior,

Port Hole :

- The chamber will be provided with 2 entry ports, each equipped with high-quality sealing gaskets/silicone plugs to ensure airtight operation.
- One port of 50 mm diameter will be positioned on the LHS of the chamber.
- One port of 100 mm diameter will be positioned on the RHS of the chamber.

Gasket:

Double-layer silicon rubber for airtight sealing.

Door Heater:

Ensures clear viewing even at temperatures 10°C lower than test conditions.

Viewing Glass:

Multilayer observation window

Size: 300 mm (W) x 400 mm (H)

Illumination: White LED lights

OPERATING ENVIRONMENT AND PERFORMANCE :

- Humidity: < 85% RH, non-condensing
- Power Supply: 3-Phase, 415V ±10%, 50Hz
- Vibration-Free Area
- No flammable or explosive gas presence

Sound Pressure Level: Noise Level: < 80 dB(A)

Electrical Panel Design :

Features:

- Separate MCBs, Contactors, and MPCBs for efficient control and protection
- Safety Controller and Safety Thermostat for enhanced safety management

Electrical Connection

Power Supply:

3 Phase + Neutral + Earth, 415V AC, 50Hz

MIL Standards:

MIL-STD-810F, MIL-STD-2164, JSS Standards: JSS 55555, JSS 6625, RTCA Standards: RTCA-DO-160E

Equivalent Indian Standards:

All applicable IS/IEC standards related to environmental simulation and vibration testing.

The chamber performance, construction, and control system are aligned to ensure compliance with these standards and equivalent updated versions as applicable.

THANK YOU!

Presented By
SESS TECHNICAL TEAM

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